

INTRO

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This is a simple applet showcasing the power of the [rumtk](#) framework. The goal of this applet is to provide a simple interface for running different kinds of benchmarks on the framework. I also hope to add an export feature to the applet so the results can be saved to disk for record keeping.

This utility is not specific to the [rumtk](#) toolset. You can configure your own profiles to examine your own programs. This benchmarking tool might not be appropriate for interpreted programs since we look at the binary's call stack. As a result, I suspect that the only useful information you might obtain from benchmarking an interpreted program is statistics about the interpreter itself.

THE BENCHMARK REPORT IS BROKEN DOWN INTO THESE SECTIONS:

- Meta: A small section showing the target architecture and OS of this rumtk-benchmark build.
- File List: list of test files and their sizes in MB used for running test of utility.
- Basic: A simple run of the selected utility using hyperfine. The resulting report shows the mean times with breakdowns by user and kernel spaces.
- Flamegraph: Interactive flamegraph attempting to visualize call graph of the test utility under load. Helpful in identifying function targets of interest for future optimizations. This visualization is generated by performing a quick benchmark with the perf command and then passing the report to [flamegraph-rs](#).
- CPU Summary: Simple summary report of basic CPU statistics obtained with the perf command.
- CPU Cache & Branch Misses: Report of functions showing a % cache and branch miss of > 1%. This helps with identifying if the performance issue is coming from a large number of CPU cache misses. Meaning, if the program is not maximizing the usage of the data already present in the CPU cache lines or a failure of the program to pre-train the CPU algorithm to prefetch necessary data. You do get a technical subreport of the current

WISHLIST:

- Network Summary: Report of the local network information.
- Network Stress Test: Test the interfaces against the message and record information about how fast everything moves through the pipes.

INFORMATION THAT IS NOT COLLECTED:

- 0 Analysis: Currently, we do not gather the information necessary to analyze the general behavior of the program or algorithm. Therefore, this benchmark utility does not provide big O notation information.
- SIMD: Currently, this utility does not inspect and present a disassembly of the test program and in particular the key functions. Thus, no information about proper usage of **SIMD** extensions can be ascertain. I recommend you use Matt Goldbolt's excellent [Compiler Explorer](#) tool or objdump for that.

CONTRIBUTING

If you appreciate the work done in this project, please, visit the following links:

- [OpenCollective](#).
- Navigate to [MedicalMasses L.L.C.'s](#) website.
- Or reach out to [Luis's email](#).

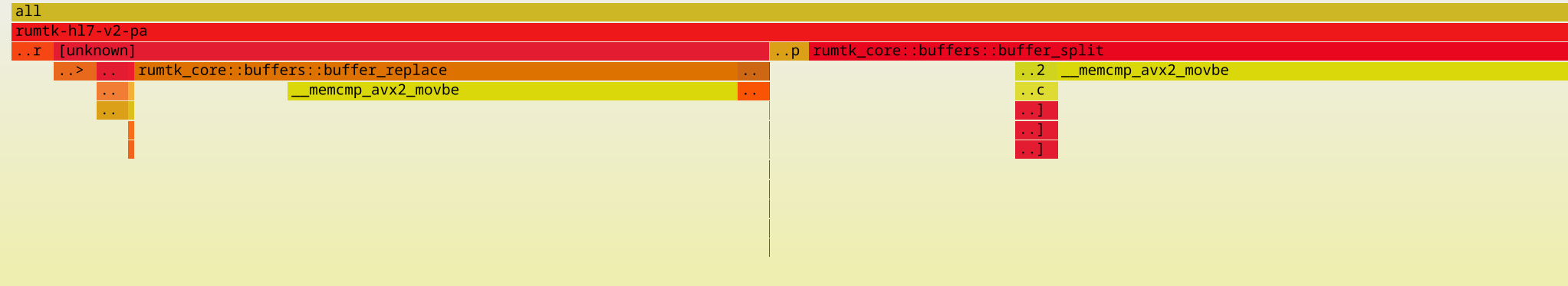
BASIC BENCHMARK

Architecture: x86_64
OS: linux

► Test File Sizes in MB

Command	Mean	Std	Min	Max	Relative	User	Kernel	Runs	Units
`../target/release/rumtk-hl7-v2-parse`	56.6	1.6	54.4	61.6	1	45.5	10.6	54	[ms]

Flame Graph



started on Fri May 8 19:50:03 2026

Performance counter stats for './target/release/rumtk-hl7-v2-parse':

2,242,090	cache-references:u
176,396	cache-misses:u
197,932,699	cycles:u
461,794,790	instructions:u
146,142,408	branches:u
5,062	faults:u
0	migrations:u

0.061478587 seconds time elapsed

0.050679000 seconds user
0.009915000 seconds sys

=====
captured on : Fri May 8 19:50:04 2026
header version : 1
data offset : 680
data size : 14384
feat offset : 15064
hostname : fedora
os release : 6.19.14-300.fc44.x86_64
perf version : 6.19.14-300.fc44.x86_64
arch : x86_64
nrcpus online : 16
nrcpus avail : 16
cpudesc : AMD Ryzen 7 7730U with Radeon Graphics
cpuid : AuthenticAMD,25,80,0
total memory : 39942832 kB
cmdline : /usr/bin/perf
event : name = cache-misses:u, , id = { 903, 904, 905, 906, 907, 908, 909, 910, 911, 912, 913, 914, 915, 916, 917, 918 }, type = 4 (cpu), size = 144, config = 0x964 (cache-misses), { sample_period, sample_freq } = 4000, sample_type = IP|TID|TIME|ID|PERIOD, read_format = TOTAL_TIME_ENABLED|TOTAL_TIME_RUNNING|ID|LOST, disabled = 1, inherit = 1, exclude_kernel = 1, exclude_hv = 1, mmap = 1, comm = 1, freq = 1, inherit_stat = 1, enable_on_exec = 1, task = 1, sample_id_all = 1, mmap2 = 1, comm_exec = 1, ksymbol = 1, bpf_event = 1, build_id = 1

```
# event : name = branch-misses:u, , id = { 919, 920, 921, 922, 923, 924, 925, 926, 927, 928,929, 930, 931, 932, 933, 934 }, type = 4 (cpu), size = 144, config = 0xc3
(ex_ret_brn_misp), { sample_period, sample_freq } = 4000, sample_type = IP|TID|TIME|ID|PERIOD, read_format = TOTAL_TIME_ENABLED|TOTAL_TIME_RUNNING|ID|LOST, disabled = 1, inherit
= 1, exclude_kernel = 1, exclude_hv = 1, freq = 1, inherit_stat = 1, enable_on_exec = 1, sample_id_all = 1
# sibling sockets : 0-15
# sibling dies      : 0-15
# sibling threads : 0-1
# sibling threads : 2-3
# sibling threads : 4-5
# sibling threads : 6-7
# sibling threads : 8-9
# sibling threads : 10-11
# sibling threads : 12-13
# sibling threads : 14-15
# CPU 0: Core ID 0, Die ID 0, Socket ID 0
# CPU 1: Core ID 0, Die ID 0, Socket ID 0
# CPU 2: Core ID 1, Die ID 0, Socket ID 0
# CPU 3: Core ID 1, Die ID 0, Socket ID 0
# CPU 4: Core ID 2, Die ID 0, Socket ID 0
# CPU 5: Core ID 2, Die ID 0, Socket ID 0
# CPU 6: Core ID 3, Die ID 0, Socket ID 0
# CPU 7: Core ID 3, Die ID 0, Socket ID 0
# CPU 8: Core ID 4, Die ID 0, Socket ID 0
# CPU 9: Core ID 4, Die ID 0, Socket ID 0
# CPU 10: Core ID 5, Die ID 0, Socket ID 0
# CPU 11: Core ID 5, Die ID 0, Socket ID 0
# CPU 12: Core ID 6, Die ID 0, Socket ID 0
# CPU 13: Core ID 6, Die ID 0, Socket ID 0
# CPU 14: Core ID 7, Die ID 0, Socket ID 0
# CPU 15: Core ID 7, Die ID 0, Socket ID 0
# node0 meminfo : total = 15444316 kB, free = 216304 kB
# node0 cpu list : 0-15
# node1 meminfo : total = 24498516 kB, free = 2341208 kB
# node1 cpu list : -1
# pmu mappings: cpu = 4, amd_df = 12, amd_iommu_0 = 14, amd_l3 = 13, breakpoint = 5, drm_amdgpu = 4294836224, hwmon_ac = 4294901760, hwmon_acpitz = 4294901761, hwmon_amdgpu =
4294901764, hwmon_bat0 = 4294901762, hwmon_k10temp = 4294901765, hwmon_nvme = 4294901763, hwmon_thinkpad = 4294901766, ibs_fetch = 10, ibs_op = 11, kprobe = 8, msr = 15, power =
16, power_core = 17, software = 1, tool = 4294967294, tracepoint = 2, uprobe = 9
# CPU cache info:
# L1 Data          32K [0-1]
# L1 Instruction    32K [0-1]
# L1 Data          32K [2-3]
# L1 Instruction    32K [2-3]
# L1 Data          32K [4-5]
# L1 Instruction    32K [4-5]
# L1 Data          32K [6-7]
# L1 Instruction    32K [6-7]
# L1 Data          32K [8-9]
# L1 Instruction    32K [8-9]
# L1 Data          32K [10-11]
# L1 Instruction    32K [10-11]
# L1 Data          32K [12-13]
# L1 Instruction    32K [12-13]
# L1 Data          32K [14-15]
# L1 Instruction    32K [14-15]
# L2 Unified       512K [0-1]
# L2 Unified       512K [2-3]
# L2 Unified       512K [4-5]
# L2 Unified       512K [6-7]
# L2 Unified       512K [8-9]
# L2 Unified       512K [10-11]
# L2 Unified       512K [12-13]
```

```
# L2 Unified 512K [14-15]
# L3 Unified 16384K [0-15]
# time of first sample : 360631.475762
# time of last sample : 360631.533812
# sample duration : 58.049 ms
# memory nodes (nr 2, block size 0x8000000):
# 0 [15G]: 198-317
# 1 [24G]: 0-25,32-197
# bpf_prog_info empty
# bt info empty
# cpu pmu capabilities: max_precise=0
# AMD systems uses ibs_op// PMU for some precise events, e.g.: cycles:p, see the 'perf list' man page for further details.
# missing features: TRACING_DATA BUILD_ID BRANCH_STACK GROUP_DESC AUXTRACE STAT CLOCKID DIR_FORMAT COMPRESSED CLOCK_DATA HYBRID_TOPOLOGY
# =====
#
#
# Total Lost Samples: 0
#
# Samples: 133 of event 'cache-misses:u'
# Event count (approx.): 194309
#
# Overhead Command Shared Object Symbol
# .....
#
# 34.13% rumtk-hl7-v2-pa /usr/lib64/libc.so.6 0x14b9e5 B [.] __memmove_avx_unaligned_erms
# 12.91% rumtk-hl7-v2-pa [unknown] 0xffffffff86a01280 ! [k] 0xffffffff86a01280
# 12.36% rumtk-hl7-v2-pa /usr/lib64/libc.so.6 0x14b064 B [.] __memcmp_avx2_movbe
# 10.55% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x117101 B [.] core::slice::memchr::memchr
# 4.76% rumtk-hl7-v2-pa /usr/lib64/libc.so.6 0x8440b B [.] cfree@GLIBC_2.2.5
# 4.13% rumtk-hl7-v2-pa /usr/lib64/libc.so.6 0x8033e B [.] _int_free_create_chunk
# 3.87% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x739fa B [.] core::ptr::drop_in_place::
# <alloc::vec::Vec<rumtk_hl7_v2::hl7_v2_parser::v2_parser::V2Field>>
# 1.98% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x7f9d3 B [.] rumtk_core::buffers::buffer_replace
# 1.91% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x7f652 B [.] rumtk_core::buffers::buffer_split
# 1.83% rumtk-hl7-v2-pa /usr/lib64/ld-linux-x86-64.so.2 0x41b6 B [.] _dl_find_object_from_map
# 1.40% rumtk-hl7-v2-pa /usr/lib64/libc.so.6 0x81541 B [.] _int_free_merge_chunk
# 1.02% rumtk-hl7-v2-pa /usr/lib64/libc.so.6 0x821b8 B [.] _int_malloc
#
# Samples: 125 of event 'branch-misses:u'
# Event count (approx.): 134614
#
# Overhead Command Shared Object Symbol
# .....
#
# 33.60% rumtk-hl7-v2-pa /usr/lib64/libc.so.6 0x14b148 B [.] __memcmp_avx2_movbe
# 12.58% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x7f657 B [.] rumtk_core::buffers::buffer_split
# 10.72% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x7b640 B [.]
# <rumtk_hl7_v2::hl7_v2_parser::v2_parser::V2Segment>::from
# 8.34% rumtk-hl7-v2-pa /usr/lib64/libc.so.6 0x82459 B [.] _int_malloc
# 5.18% rumtk-hl7-v2-pa [unknown] 0xffffffff8812c8c6 ! [k] 0xffffffff8812c8c6
# 4.83% rumtk-hl7-v2-pa /usr/lib64/libc.so.6 0x803d8 B [.] _int_free_create_chunk
# 4.39% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0xed3f0 B [.] __rustc::rdl_alloc
# 2.89% rumtk-hl7-v2-pa /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x73a02 B [.] core::ptr::drop_in_place::
# <alloc::vec::Vec<rumtk_hl7_v2::hl7_v2_parser::v2_parser::V2Field>>
# 2.65% rumtk-hl7-v2-pa /usr/lib64/libc.so.6 0x8440b B [.] cfree@GLIBC_2.2.5
```

```
2.51%  rumtk-hl7-v2-pa  /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x730e7 B [...] <&mut serde_json::ser::Serializer<&mut
alloc::vec::Vec<u8>, serde_json::ser::PrettyFormatter> as serde_core::ser::Serializer>::serialize_str
2.23%  rumtk-hl7-v2-pa  /usr/lib64/libc.so.6 0x82f98 B [...] __libc_malloc2
1.62%  rumtk-hl7-v2-pa  /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x7f455 B [...] rumtk_core::buffers::buffer_trim
1.55%  rumtk-hl7-v2-pa  /usr/lib64/libc.so.6 0x81570 B [...] _int_free_merge_chunk
1.18%  rumtk-hl7-v2-pa  /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x7adb8 B [...]
<rumtk_hl7_v2::hl7_v2_parser::v2_parser::V2Field>::from
1.14%  rumtk-hl7-v2-pa  [unknown] 0xffffffff86a01280 ! [k] 0xffffffff86a01280
1.14%  rumtk-hl7-v2-pa  /home/kiseitai2/RustroverProjects/rumtk/target/release/rumtk-hl7-v2-parse 0x7f884 B [...] rumtk_core::buffers::buffer_replace

#
# (Tip: To show time in nanoseconds in record/report add --ns)
#
```